

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester V Examination, 2016-17

Paper No. GLB-502: Igneous Petrology, Mineralogy and Crystallography

Time: 3 Hours

Maximum Marks: 70

Ref. No.: BS/Sem V/73

*Instructions: Attempt any **five** questions, selecting at least **two** from each of Sections A and B. Section C is compulsory. All questions carry equal marks (14 marks each).*

SECTION — A

Question 1. Give the salient features of chemical classification schemes for igneous rocks. [14]

Question 2. Write the petrographic description of **any four** of the following: $[4 \times 3.5 = 14]$

- (a) Norite
- (b) Komatiite
- (c) Rhyolite
- (d) Andesite
- (e) Dunite
- (f) Phonolite

Question 3. Write descriptive notes on the following: $[2 \times 7 = 14]$

- (a) Fractional crystallization
- (b) Types of magmas

Question 4. Describe with the help of a neat sketch the crystallization of magma in the phase equilibrium of Diopside-Albite-Anorthite system and comment on its petrogenetic significance. [14]

SECTION — B

Question 5. Describe the symmetry and forms belonging to the holosymmetric class of the Monoclinic system and draw a suitable stereogram. Name any four minerals belonging to this class. [14]

Question 6. Write short notes on **any two** of the following: $[2 \times 7 = 14]$

- (a) Optic sign

- (b) Solid solution
- (c) Plagioclase feldspars

Question 7. Discuss in detail the structural, physical, chemical, and optical properties of the mica group of minerals. [14]

SECTION — C (Compulsory)

Question 8. Answer the following questions (each carry 2 marks): [7 × 2 = 14]

- (i) Name any two igneous rocks which are ultramafic as well as ultrabasic.
- (ii) What are the factors that control the stability of minerals?
- (iii) What is the utility of the Michel-Lévy Chart?
- (iv) How do we know that there is water in the Earth's mantle?
- (v) What is the difference between eutectic and cotectic?
- (vi) How many optic axes are there in minerals crystallizing in the Triclinic system?
- (vii) What is the extinction angle in quartz?